

DECLARATION

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This is to certify that the thesis work entitled "(Name of Thesis Title)" has been carried out by (Name of the Student) under the supervision of (Name of the Supervisor) in the Department of Mechatronics Engineering, Khulna University of Engineering & Technology, Khulna, Bangladesh. The above thesis work or any part of this work has not been submitted anywhere for the award of any degree.

The above declarations are true. Understanding these, this work has been submitted for the evaluation of an undergraduate thesis.

Signature of Supervisor

Signature of Candidate

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Department of Mechatronics Engineering

KUET, Khulna-9203, Bangladesh.

Date: (17 February, 2024)

APPROVAL

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This is to certify that the thesis work submitted by (Name of the Student) entitled " (Name of Thesis Title)" has been approved by the board of examiners for the partial fulfillment of the requirements for the degree of Bachelor of Science in the Department of Mechatronics Engineering, Khulna University of Engineering & Technology, Khulna, Bangladesh in February 2019.

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ABSTRACT

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TABLE OF CONTENTS

(Font size 14pt, Bold, Times New Roman and Centered)

One blank line

[Times New Roman font 12pt, Table title bold and body normal font]

(Hide the table after editing)

		Page
Acknowledgment		iv
Abstract		v
Table of Contents		vi
List of Tables		viii
List of Figures		ix
List of Nomenclature		x
CHAPTER 1	Introduction	1
	1.1 General	1
	1.2 Scope of present Investigation	4
	1.3 Project report layout	5
CHAPTER 2	Motivation/ Background Study	6
	2.1 Subtopic 1	6
	2.2 Subtopic 2	8
	2.3 Subtopic 3	10
	2.3.1 Subtopic 1	10
	2.3.2 Subtopic 2	10
	2.3.3 Subtopic 3	11
	2.4 Subtopic 4	15
	2.5 Subtopic 5 etc.	16
CHAPTER 3	Methodology	24
	3.1 Subtopic 1	24
	3.2 Subtopic 2	25
	3.2.1 Subtopic 1	25

	3.2.2 Subtopic 2	26
	3.3 Subtopic 3 etc	31
CHAPTER 4	Design procedure/Experimental set-up/ Circuit Diagram	35
	4.1 Introduction	35
	4.2 Materials and methods	35
	4.3 Experimental Set-up	35
	4.4 Experimental Procedure	39
	4.5 Experimental Data	40
CHAPTER 5	Implementation	45
	5.1 Subtopic 1	45
	5.2 Subtopic 2	45
	5.3. Subtopic 3 etc	49
CHAPTER 6	Results and Discussions	69
	6.1 Results	69
	6.2 Discussions	70
CHAPTER 7	Conclusion and Future Work	70
	7.1 Conclusion	
	7.2 Future Work	70
	References	71
	Appendix A	72
	Appendix B etc.	73

LIST OF TABLES

(Font size 14pt, Bold, Times New Roman and Centered)

One blank line

[Times new roman font 12pt, Table title bold and body normal font]

(Hide the table after editing)

Table No.	Description	Page
2.1		8
2.2		22
4.1		40
4.2		41
4.3		41
4.4		42
4.5		42
4.6		43
4.7		43
4.8		44
4.9		44
4.10		44
5.1		46
5.2		46
5.3		46

LIST OF FIGURES

(Font size 14pt, Bold, Times New Roman and Centered)

One blank line

[Times new roman font 12pt, Table title bold and body normal font]

(Hide the table after editing)

Figure No.	Description	Page
2.1		6
2.2		14
2.3		15
3.1		26
4.1		36
4.2		37
4.3		37
4.4		38
4.5		38
5.1		49
5.2		50
5.3		50
5.4		51
5.5		51
5.6		52
5.7		52
5.8		53
5.9		53

List of Nomenclature

(Font size 14pt, Bold, Times New Roman and Centered)

One blank line

[Times New Roman font 12pt normal font]

(Hide the table after editing)

wt %	Weight Percent
°C	Degree Celsius
MJ/kg	Mega joule per Kilogram
Kg/m ³	Kilogram per cubic meter
KgOE	Kilogram of Oil Equivalent
GCV	Gross Calorific value
SBR	Styrene Butadiene Rubber
PVC	Poly Vinyl Chloride
PET	Poly Ethelene Terapthalate
PS	Poly Styrene
EPT	Ethylene Propylene Terpolymer
IIR	Isobutylene Iso-prene Rubber
UV	Ultraviolet
NR	Natural Rubber
EPA	European Pollution Authority
MJ	Mega Joule
FCC	Fluid Catalytic Cracking
TGA	Thermo Gravimetric Analysis
KW	Kilowatt
N ₂	Nitrogen Gas
O-H	Hydroxyl stretching
N-H	Amines
C-H	Carbon Hydrogen (Alkanes group)
C=C	Carbon Hydrogen double bonding
C-O	Carbonyl Stretching
O-H	Alcohol, Phenol group
C≡C	Carbon Hydrogen Triple Bonding (Alkynes)